Timeline Check in

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2026-05-22

Ideas for analysis

Questions

- How does the percent cover of native grasses vary over time compared to the percent cover of exotic vegetation on the NCOS Mesa Perennial Grasslands?
  - How does intentional disturbance influence this relationship?
  - How do key individual species, including Stipa pulchra, Medicago polymorpha, Melilotus indicus, and Festuca perennis, vary in percent cover before and after intentional disturbance?

Statistical analysis

- statistical analysis to address at least one question with at least one analysis coded and run

```r
# run paired t-test for each species and extract key statistics
focal_t_test_results <- focal_change_transect_pp |>
  # group by spp and spp label
  group_by(psoc, species_label) |>
  # calculate and store full t-test output as a list column
  summarize(
    # year_2023 and year 2024 are the mean percent cover values from
    ↪  respective years
    t_test = list(t.test(year_2023, year_2024, paired = TRUE))
  ) |>
  mutate(
    # extract t-statistic from each t-test result
```

```r
    t_stat = map_dbl(t_test, ~ .x$statistic),
    # extract degrees of freedom from each t-test result
    df = map_dbl(t_test, ~ .x$parameter),
    # extract p-value from each t-test result
    p_value = map_dbl(t_test, ~ .x$p.value)
  ) |>
  # removing groups
  ungroup() |>
  # drop psoc and t_test list column; keep only display columns
  select( species_label, t_stat, df, p_value)

# format results as a flextable for report output
focal_t_test_table <- focal_t_test_results |>
  flextable() |>
  # rename columns for readability
  set_header_labels(
    species_label = "Species",
    t_stat = "t-statistic",
    df = "df",
    p_value = "p-value"
  ) |>
  # round t-statistic and p-value to 3 decimal places
  colformat_double(j = c("t_stat", "p_value"), digits = 3) |>
  # italicize species names
  italic(j = "species_label") |>
  # bold significant p-values
  bold(i = ~ p_value < 0.05, j = "p_value") |>
  # auto fit column widths
  autofit()

focal_t_test_table <- bg(focal_t_test_table, bg = "white", part = "all")

# display table
focal_t_test_table
```

Table 1: Paired t-test results comparing mean percent cover of four focal species before (2023) and after (2024) the prescribed burn along Mesa (GL) transects at NCOS (n = 7 transects). Significant p-values ($p < 0.05$) are bolded. Tests were conducted without correction for multiple comparisons.

Species	t-statistic	df	p-value
F. perennis	1.013	6	0.350
M. polymorpha	0.381	6	0.716
M. indicus	-3.508	6	**0.013**
S. pulchra	4.658	6	**0.003**

Background

We assess whether the 2023 cultural burn is meeting its restoration goals of promoting native grassland species. This study may help to inform future restoration management decisions regarding intentional disturbance at NCOS and could have implications which extend to the management or ecological understanding of other coastal grassland systems. These questions are relevant because Santa Barbara coastal grasslands are fire adapted (Saglimbeni, 2024, pp. 1-2), but are now heavily affected by invasive species. Studying the NCOS burn helps show whether cultural burning can support native species like Stipa pulchra while reducing nonnative grassland invaders. The four focal species, Stipa pulchra, Medicago polymorpha, Melilotus indicus, and Festuca perennis, were selected because a prior study found these native and non-native species showed contrasting abundance responses to disturbance, making them useful indicators for evaluating changes in grassland composition (Nestor, 2025).

Visualizations

Plot 1: Native and exotic percent cover over time

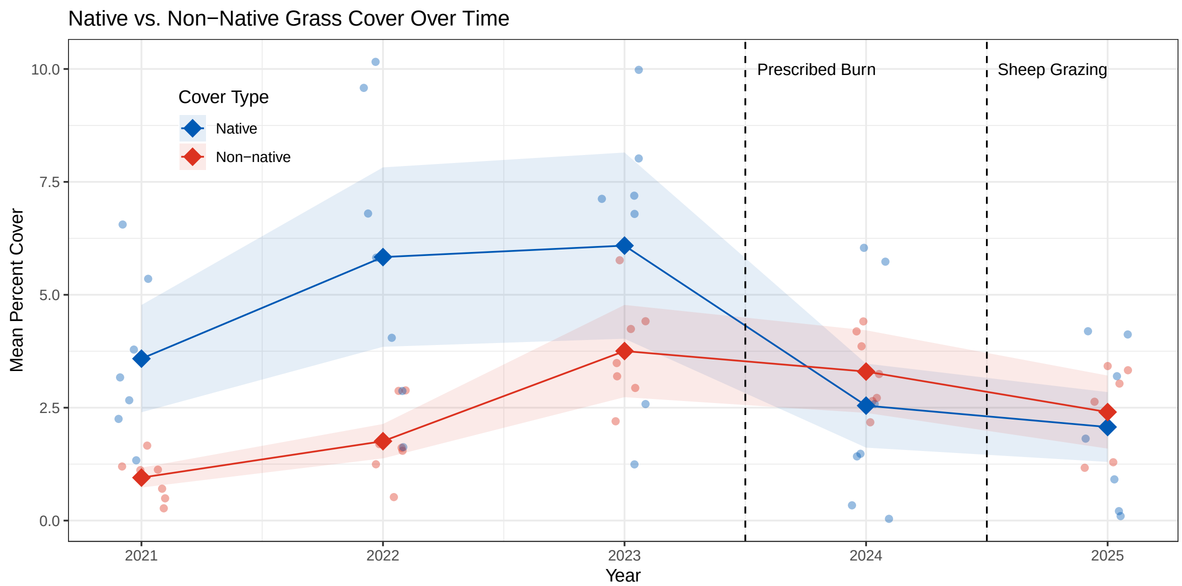

Figure 1: Mean percent cover of native and non-native grasses along the Mesa (GL) transects at the North Campus Open Space (NCOS) from 2021 to 2025. Filled circles represent the grand mean across transects and open circles represent individual transect means. Shaded ribbons show ±1 standard error. Dashed vertical lines indicate the prescribed burn (Fall 2023) and rotational sheep grazing (Fall 2024) treatment events.

Plot 2: Sweet and bur clover abundance over time

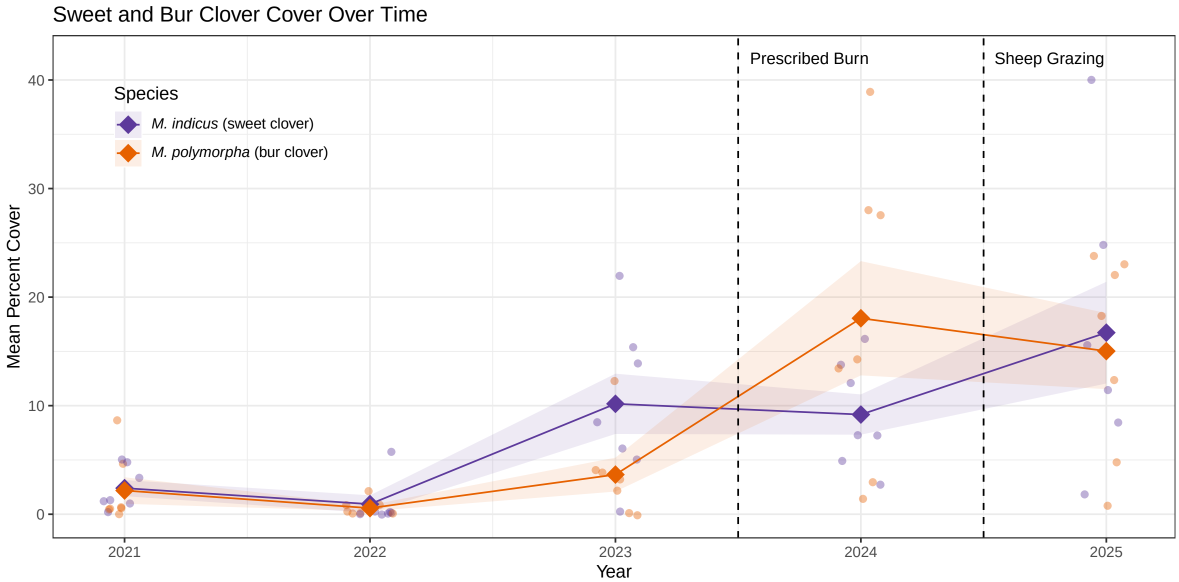

Figure 2: Mean percent cover of bur clover (M. polymorpha) and sweet clover (M. indicus) along the Mesa (GL) transects at the North Campus Open Space (NCOS) from 2021 to 2025. Filled circles represent the grand mean across transects and open circles represent individual transect means. Shaded ribbons show ±1 standard error. Dashed vertical lines indicate the prescribed burn (Fall 2023) and rotational sheep grazing (Fall 2024) treatment events. Field observations suggested M. polymorpha increased following the prescribed burn and M. indicus following sheep grazing, a pattern reflected in the monitoring data.

Plot 3: Change in mean percent cover of key species after disturbance

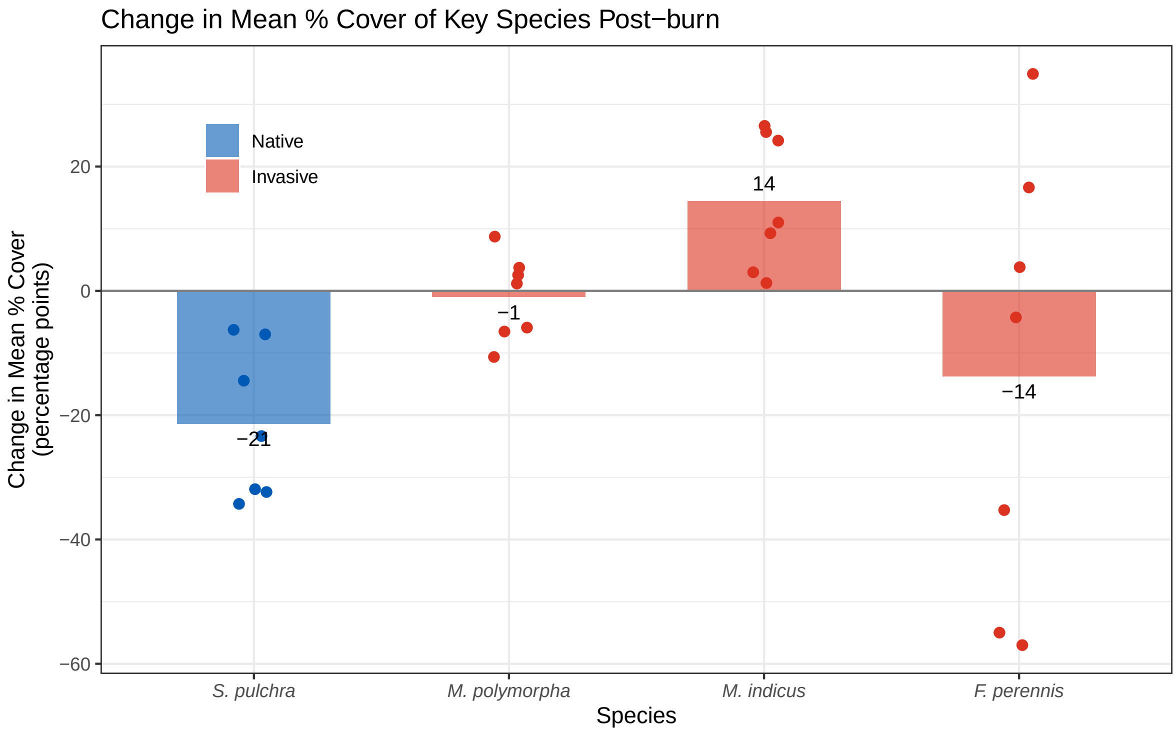

Figure 3: Change in mean percent cover of four key grassland species on the NCOS Mesa from 2023, pre-burn, to 2024, one year post-burn. Bars represent the average percent-point change in cover across transects for each species, while individual points show transect-level percent-point changes; the gray horizontal line represents no change, with values above zero indicating increased cover and values below zero indicating decreased cover. Blue indicates the native species, Stipa pulchra, and red indicates invasive species: Medicago polymorpha, Melilotus indicus, and Festuca perennis. Overall, S. pulchra and F. perennis decreased after the burn, M. indicus increased, and M. polymorpha showed little average change. This figure connects to our main question by showing that intentional disturbance did not affect all native and invasive species in the same way.

Plot 4: Mean percent cover of native grasses after disturbance

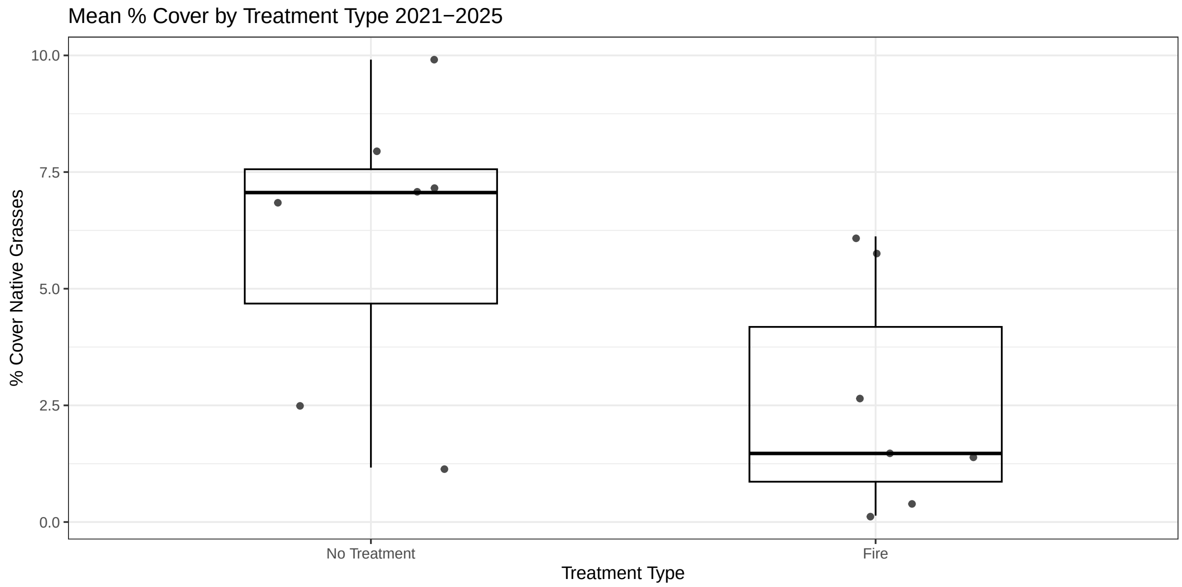

Figure 4: Change in native mean % cover by treatment type from 2021-2024. Bars represent the average percent-point change in cover across transects for all native species. Individual points show transect level percent points; Black line represents average mean percent cover. Yellow bar indicates no treatment while orange bar indicates fire treatment. From the graph, it shows that after a disturbance like fire, native percent cover for grasses decreases at least one year post treatment. This figure connects to our main question by showing that intentional disturbance did not affect all native and invasive species in the same way.

Exploratory Visualization 1: Change in mean percent cover of key species after sheep grazing

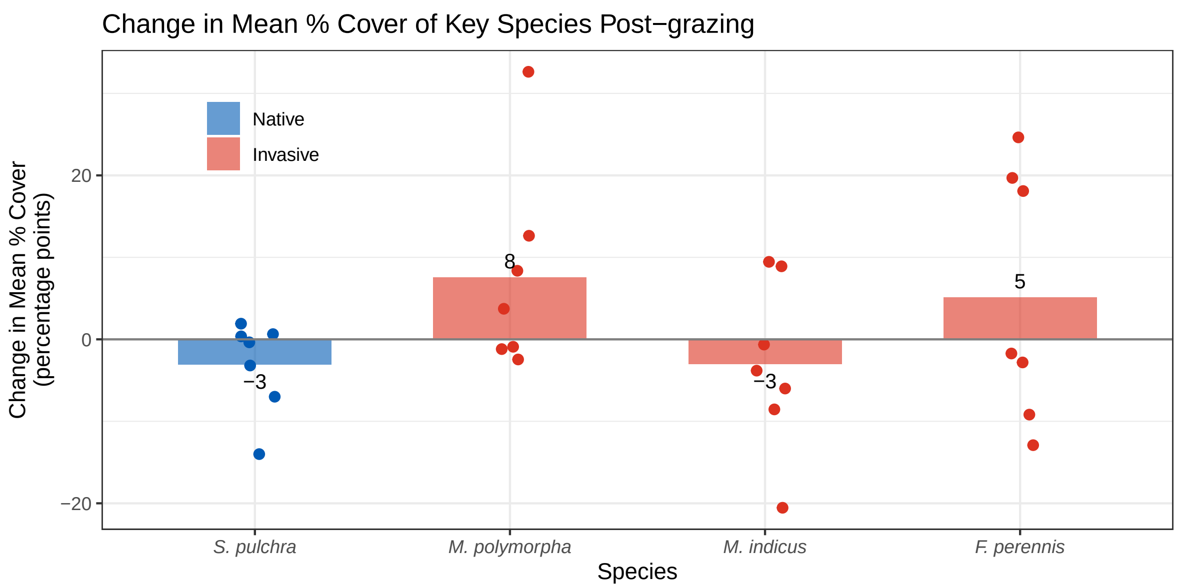

Figure 5: Change in mean percent cover of four key grassland species on the NCOS Mesa from 2024, pre-grazing, to 2025, post-grazing. Bars show the average percent-point change across transects, individual points show transect-level changes, and the gray horizontal line represents no change; blue indicates the native species Stipa pulchra, while red indicates invasive species. As an exploratory visualization, this figure suggests that S. pulchra changed only slightly after grazing, while invasive species showed mixed responses, with M. polymorpha and F. perennis increasing on average and M. indicus decreasing slightly.

Exploratory Visualization 2: Effect of annual precipitation on native and invasive percent cover

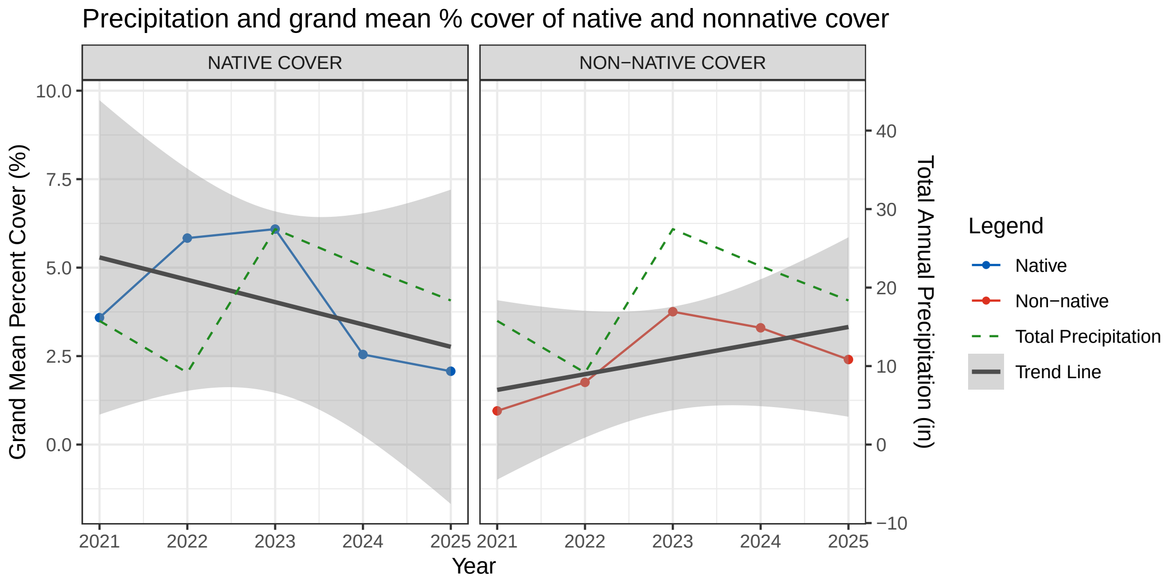

Figure 6: Changes in grand mean percent cover for native and non-native cover categories on NCOS Mesa from 2021-2025. Individual points show grand mean percent cover for each year. Dark grey line shows trendline; blue line represents native cover; orange represents non-native cover; dotted line represents total annual precipitation. As an exploratory visualization, this figure suggests that precipitation is a factor that increases non-native percent cover while precipitation doesn't increase native percent cover.

Plan for elective

- The general concept is we are making an ArcGIS StoryMap presenting our NCOS Mesa vegetation analysis. The StoryMap walks through site context, the 2023 prescribed burn, methods, results, and an embedded interactive dashboard where users can explore transect photo points and a spatial percent cover heatmap across the Mesa Perennial Grassland.
- The full structure is built out and mostly populated. The completed sections include site context, burn background, methods, and results with all four figures and the t-test table embedded. Remaining work includes building the ArcGIS Dashboard with photo point popups and the interpolated heatmap layer, embedding it into the StoryMap, and finalizing the discussion and references sections.

- Link to StoryMap